# Girls Build Bots — Brownie STEM Career Exploration

## Badge-aligned workshop guide for robotics-team-led outreach

**Level:** Brownies (Grades 2–3) · **Badge:** STEM Career Exploration **Format:** One 75–90 minute session · **Ratio:** 1 mentor per 3–4 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. This badge is the one where **the team itself is the main resource** — every mentor is a living example of a STEM path in progress. A robot demo is a wonderful bonus but not required.

> **Troop Leaders:** This curriculum is designed to *align with* the Brownie STEM Career Exploration badge. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

---

## The big idea

STEM careers aren't far away and grown-up — they grow out of things Brownies *already love*. A Scout who likes animals could be a wildlife vet or a marine biologist; one who likes building could be an engineer; one who likes drawing could design video games. The goal: connect what she loves to a real STEM job, and meet people (the team!) on that path.

**Vocabulary:** STEM (science, technology, engineering, math), career, engineer, scientist, designer, mentor.

---

## STEM Career Exploration (75–90 minutes)

### Materials

- "STEM career" picture cards (vet, video-game designer, roller-coaster engineer, astronaut, robot builder, marine biologist, architect)
- "What I love" sheet: things I like to do → a STEM job that uses it
- The robotics team members themselves, ready to share short "my STEM story" moments

### Activity 1 — What Do You Love? (10 min)

Go around: everyone names one thing she loves to do (animals, drawing, building, space, puzzles). Write them up. No wrong answers — this is the raw material for the whole session.

### Activity 2 — Meet Real STEM People: The Team Panel (25 min)

The heart of the badge. 3–4 team members each give a **2-minute "my STEM story"**: what they build/do, how they got started, and one thing they love about it. Keep it warm and real

(including the hard-but-fun parts). Then Scouts ask questions. Seeing near-peer role models — especially young women in robotics — is the single most powerful part of this badge.

### Activity 3 — Match Your Love to a Career (25 min)

Using the picture cards, Scouts match things they love to STEM jobs that use them (love animals → marine biologist who studies dolphins with underwater robots; love games → game designer). Each Scout fills her "What I love → STEM job" sheet for at least one match. Mentors help make surprising connections.

### Activity 4 — My STEM Dream + Wrap (15 min)

Each Scout shares one STEM job she'd like to explore and **one small step** she could take (read a book about it, try a coding game, ask a question). Optional: a quick robot demo to end on a high note. Closing: everyone says "I could be a ___."

> **Robot moment (optional):** If you brought a robot, let a few Scouts drive it or trigger it — nothing makes "I could do this" feel real like touching the machine.

### Completion Checklist — STEM Career Exploration

By the end of the session, to earn the badge, each Scout can: - [ ] Name what STEM stands for (science, technology, engineering, math) - [ ] Describe a STEM career after meeting someone on that path - [ ] Connect something she loves to a real STEM job - [ ] Name one STEM career she'd like to explore and one step to take

---

## Mentor Guide (for the robotics team)

- **You are the curriculum.** Your stories matter more than any worksheet. Prepare your 2-minute "my STEM story" in advance — beginning, a struggle, why you love it.
- **Near-peer is the magic.** You're only a few years older than these Scouts. Say "I started right where you are." It lands harder than any adult could.
- **Center the girls on your team.** If you have young women in technical roles, make sure Scouts hear from them — representation is the point.
- **Connect, don't correct.** When a Scout says she loves something "non-STEM," find the STEM in it (loves cooking → food scientist; loves fashion → materials engineer).
- **End with a step.** Every Scout should leave with one concrete, doable next action.

---

## 📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.

2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

---

*Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Brownie STEM Career Exploration badge; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.*